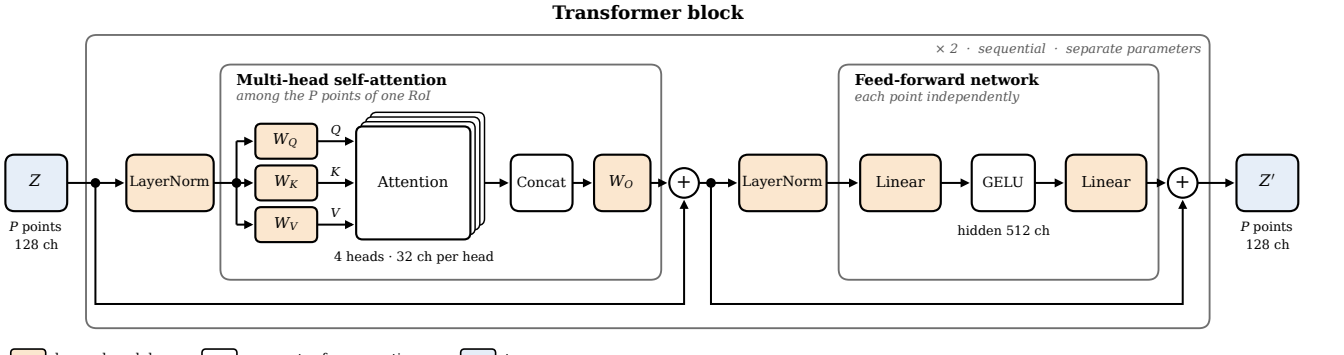
Transformer block
× 2 · sequential · separate parameters
Multi-head self-attention
among the P points of one RoI
Feed-forward network
each point independently
Z
P points
128 ch
LayerNorm
W_Q
W_K
W_V
Q
K
V
Attention
4 heads · 32 ch per head
Concat
W_O
LayerNorm
Linear
GELU
Linear
hidden 512 ch
Z′
P points
128 ch
learned module
parameter-free operation
tensor